

Multiple Choice (10 marks)

1. Which of the following is a primary standard for use in standardizing bases?
 - (a) Ammonium hydroxide
 - (b) Sulfuric acid
 - (c) Acetic acid
 - (d) Potassium hydrogen phthalate
2. Which of the following experimental observations were explained by Planck's quantum theory?
 - (a) Blackbody radiation curves
 - (b) Emission spectra of diatomic molecules
 - (c) Electron diffraction patterns
 - (d) Temperature dependence of reaction rates
3. The anhydride of $\text{Ba}(\text{OH})_2$ is
 - (a) BaH_2
 - (b) BaOH
 - (c) Ba
 - (d) BaO
4. The T_2 of an NMR line is 15 ms. Ignoring other sources of linebroadening, calculate its linewidth.
 - (a) 0.0471 Hz
 - (b) 21.2 Hz
 - (c) 42.4 Hz
 - (d) 66.7 Hz
5. Which of the following metal ions cannot be used as a paramagnetic quencher?
 - (a) Ti_3^+
 - (b) Cr_3^+
 - (c) Fe_3^+
 - (d) Zn_2^+

True / False (10 marks)

Answer True or False

6. True or False: H_3O^+ and H_2O form a conjugate acid-base pair in the Brønsted-Lowry acid-base theory.
7. True or False: MgCl_2 is less likely to behave as a Lewis acid than SCl_2 because it has a full octet.
8. True or False: The solid-state structures of the principal allotropes of elemental boron are composed of B_{12} icosahedra.
9. True or False: The ^{13}C spectrum of hexane has five distinct chemical shifts because of its symmetrical structure.
10. True or False: The limiting high-temperature molar heat capacity at constant volume (C_V) of a gas-phase diatomic molecule is $2.5 R$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the trends in atomic radii among the lanthanide elements and analyze why the statement regarding their increase from La to Lu is incorrect.
12. Discuss the role of entropy in spontaneous processes and explain how the total entropy of a system and its surroundings changes during such processes.

End of Paper